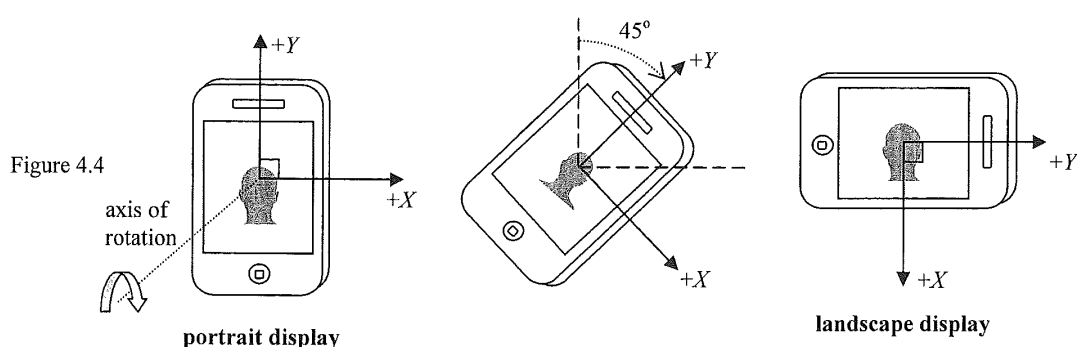


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Electronic accelerometers employing similar principles are widely used in smart phones. The phone's orientation is detected by a set of built-in accelerometers each detecting the acceleration due to gravity along mutually perpendicular axes of the phone. A phone in vertical orientation is shown on the left of Figure 4.4, its accelerometer along the Y axis would be sensing the acceleration due to gravity, denoted by $a_Y = -g$. When the phone is rotated about a horizontal axis perpendicular to both X and Y axes by more than 45° , 'portrait display' would change to 'landscape display' as shown on the right of Figure 4.4.



- (c) What would the kind of display be if the phone is rotated clockwise until the acceleration a_Y sensed by the accelerometer along the rotated Y axis is $-0.5 g$? Explain. (2 marks)

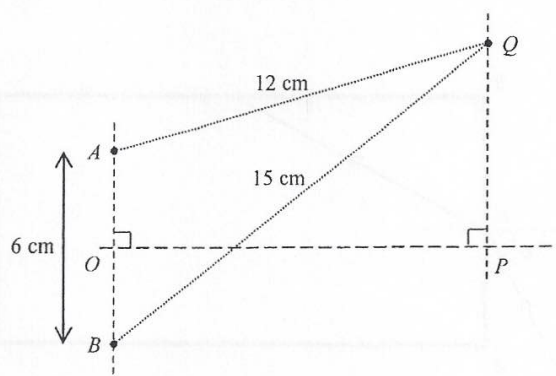
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- (b) In Figure 6.3, A and B are two dippers vibrating in phase in a water tank. The distance between A and B is 6 cm. OP is the perpendicular bisector of AB . Q is a second minimum from P , where $AQ = 12$ cm and $BQ = 15$ cm.

Figure 6.3

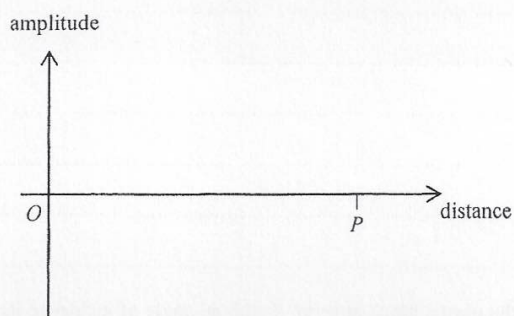


- (i) Explain why a minimum occurs at Q . (2 marks)

- (ii) Determine the wavelength of the water wave. (2 marks)

- (iii) Sketch in Figure 6.4 how the **AMPLITUDE** of the water wave varies along the line OP . (1 mark)

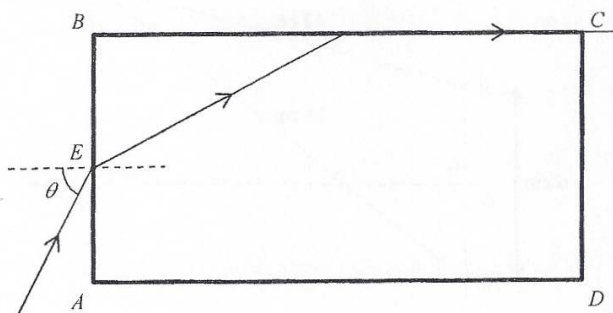
Figure 6.4



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7. (a) A light ray enters a rectangular plastic block $ABCD$ from air at point E , and the angle of incidence is θ . The light ray emerges along face BC as shown in Figure 7.1. The refractive index of the plastic is 1.36.

Figure 7.1



- (i) Find the critical angle of the plastic. (2 marks)

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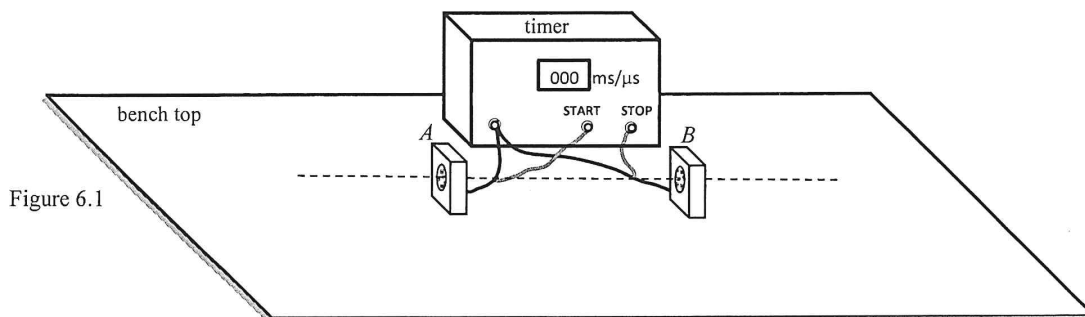
- (ii) Find the value of θ . (3 marks)

- (iii) If the light ray enters the plastic block at point E with an angle of incidence larger than θ , sketch the path of the light ray in Figure 7.1. (2 marks)

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6. The set-up in Figure 6.1 is to find the speed of sound in air. Two identical microphones *A* and *B* are connected to a timer and placed on a bench top as shown. The timer can be triggered to 'start' and 'stop' timing using the respective microphones to feed signals to the START and STOP terminals of the timer.



- (a) You are given a hammer and a metal plate (). Use 'X' to indicate a suitable location on Figure 6.1 where the hammer should hit the plate so as to generate a sharp loud sound to be received by the microphones in this experiment. State an additional piece of apparatus needed and the measurements to be made in this experiment. (3 marks)
- (b) The separation between *A* and *B* is set at 0.280 m. The experiment is repeated to obtain a few readings of the timer as follows:

801 μs , 838 μs , 539 μs , 821 μs

- (i) Find the speed of sound in air. Show how you would treat the data obtained in the calculation.
- (ii) Suggest one adjustment to the experimental setting so as to obtain a more accurate result. (3 marks)

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7. Figure 7.1 shows an optical fibre which consists of a cylindrical glass core of refractive index n_g enclosed by a transparent cladding of refractive index n_c .

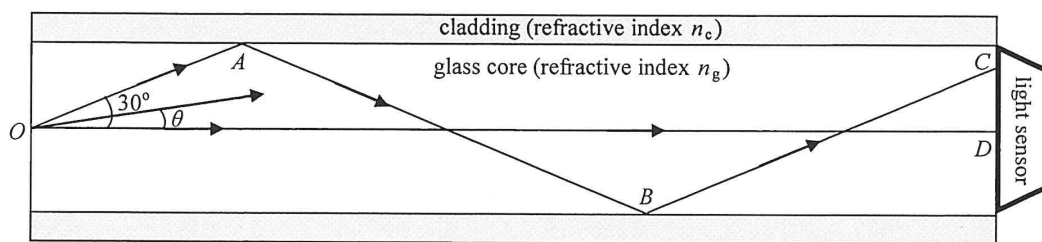


Figure 7.1

As shown in Figure 7.1, a point light source at O emits monochromatic light in all directions. Inside the fibre, light can reach the right end of the fibre through many different paths making angles θ with the axis OD . Two of these paths, OD and $OABC$, have been drawn for reference. Light ray OA makes an angle of 30° with the axis OD and is incident at the core-cladding boundary at A with an angle of incidence i_A .

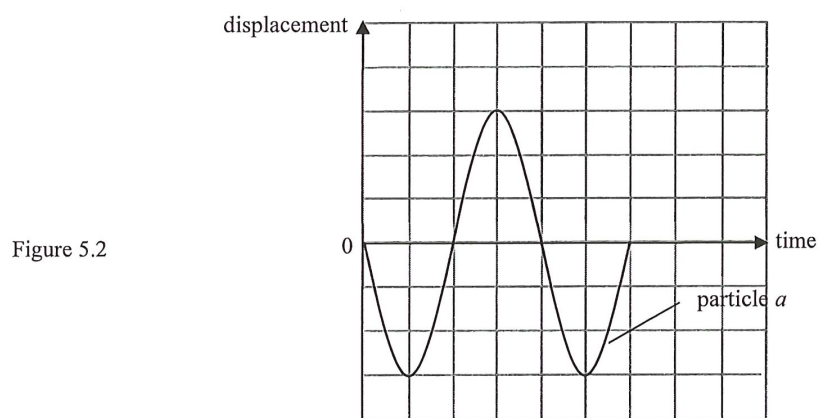
- (a) (i) Find i_A . (1 mark)

- (ii) If i_A is just greater than the critical angle of that boundary, estimate $\frac{n_g}{n_c}$. (2 marks)

- (iii) What phenomenon occurs at point A ? State the condition needs to be satisfied by θ such that this phenomenon **fails to occur**. (2 marks)

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- (iii) Figure 5.2 shows the displacement-time graph of particle a . Sketch on the same figure the corresponding graph for particle m . (2 marks)



- (c) State the change on the position of particle h at time $t = 0$ if the loudspeaker is turned louder. (1 mark)

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6. Read the following passage about **rainbows** and answer the questions that follow.

Rainbows are sometimes seen after a rainfall (Figure 6.1(a)). Tiny water droplets in the air each act like a prism to disperse sunlight into a spectrum of colours. The most common rainbow that we see is called a primary rainbow, in which the white light from the sun undergoes reflection once within each water droplet.



Figure 6.1(a)

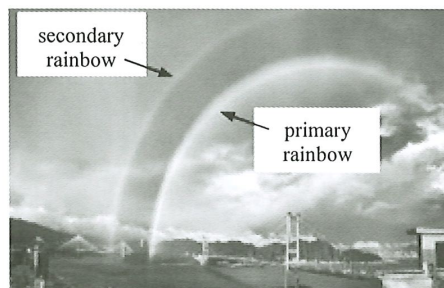


Figure 6.1(b)

Under suitable conditions, a double rainbow with an additional secondary rainbow formed above the primary rainbow can be seen (Figure 6.1(b)).

Assume that water droplets are spherical. Given: refractive index for red light in water = 1.325

(a) Figure 6.2 shows a red light ray entering a water droplet suspended in air.

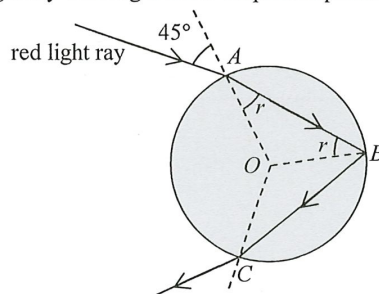


Figure 6.2

(i) When the angle of incidence is 45° , find the angle of refraction r at point A . (2 marks)

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(ii) Calculate the critical angle c for red light at the water-air interface and justify whether the reflection at point B is a total internal reflection or not. (3 marks)

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How does the temperature difference between air layers above the ground explain the situation in Figure 7.2 ? (2 marks)

(c) It is known that the air temperature is 10°C . Referring to the graph in Figure 7.1:

(i) estimate the ratio of speed of light to the speed of sound; (1 mark)

(ii) estimate the distance of a thunderstorm from an observer O if thunder is heard 3.0 s after lightning is seen by O . (2 marks)

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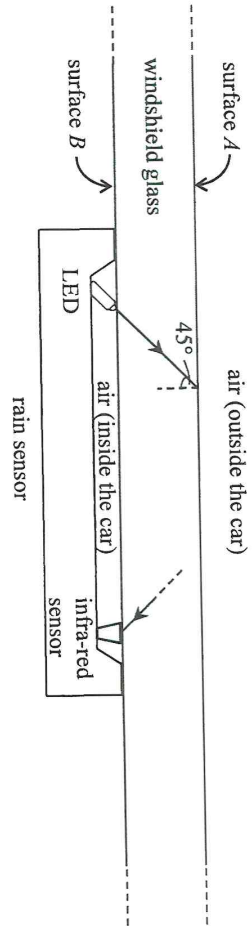
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As shown in Figure 10.2, the rain sensor is composed of a light emitting diode (LED) and an infra-red sensor. The LED emits an infra-red signal and the infra-red sensor measures the intensity of the signal. When the intensity changes, the wipers are activated. An infra-red ray from the LED incident at surface A at 45° , and the infra-red ray finally received by the infra-red sensor are drawn in Figure 10.2. Assume surface A is parallel to surface B . It is given that the critical angle of the glass for infra-red is $44,0^\circ$.

Figure 10.2



- (a) Explain how the infra-red ray in Figure 10.2 travels from the LED to the infra-red sensor. (4 marks)
